

You've completed quite a lot already:

- ☐ Flask Basics
 - ☐ SQLAlchemy Models
 - ☐ PostgreSQL
 - ☐ CRUD APIs
 - ☐ Service Layer
 - ☐ Repository Pattern
 - ☐ JWT Authentication
 - ☐ Access + Refresh Tokens
 - ☐ Marshmallow Validation
 - ☐ Custom Exceptions
 - ☐ Global Error Handling
 - ☐ Logging
 - ☐ Pagination
 - ☐ RBAC
 - ☐ Redis Caching
 - ☐ Rate Limiting
 - ☐ Postman Documentation
 - ☐ Git & GitHub
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Phase 2: Professional Backend

API & Documentation

1. Swagger / OpenAPI
 2. API Versioning
 3. Standardized API Responses
 4. DTO Pattern
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Security

5. Email Verification
 6. Forgot Password Flow
 7. Password Reset Tokens
 8. JWT Blacklisting (Logout)
 9. CORS
 10. Security Headers
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Testing

- 11. Pytest Basics
 - 12. Unit Testing
 - 13. Integration Testing
 - 14. Mocking
 - 15. Coverage Reports
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Background Processing

- 16. Celery
 - 17. Redis Queues
 - 18. Scheduled Jobs
 - 19. Email Queue
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File Management

- 20. File Uploads
 - 21. Image Uploads
 - 22. AWS S3 Storage
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DevOps

- 23. Docker
 - 24. Docker Compose
 - 25. Environment Config (Dev/Prod)
 - 26. Nginx
 - 27. Linux Deployment
 - 28. AWS EC2 Deployment
 - 29. CI/CD Pipeline (GitHub Actions)
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Database Advanced

- 30. Database Indexing
- 31. Query Optimization
- 32. Transactions

- 33. Alembic Advanced
 - 34. Database Seeding
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Messaging

- 35. RabbitMQ
 - 36. Kafka Basics
 - 37. Event-Driven Architecture
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System Design

- 38. Load Balancing
 - 39. Horizontal Scaling
 - 40. Database Replication
 - 41. Caching Strategies
 - 42. CAP Theorem
 - 43. Microservices Basics
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Architecture

- 44. Clean Architecture
 - 45. Dependency Injection
 - 46. DDD Basics
 - 47. CQRS Basics
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AI Backend (Very Relevant for You)

- 48. Vector Databases
- 49. RAG APIs
- 50. Streaming APIs
- 51. WebSockets
- 52. SSE (Server Sent Events)
- 53. LangGraph Backend Architecture
- 54. Multi-Agent Backend Design
- 55. MCP Servers
- 56. AI Model Serving

